Assessing the use of Physics and AI-Driven Precipitation Forecasts for Streamflow Prediction Using the NGIAB

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Introduction

Accurate streamflow forecasting is critical for water resource management, flood prediction, and disaster preparedness in densely populated regions such as the Hackensack River watershed within the New York City Metropolitan Area. Models like the NextGen In A Box (NGIAB) framework, National Water Model (NWM) depend on hydrological precipitation inputs from regional numerical weather prediction (NWP) systems, including the Weather Research and Forecasting (WRF) model and the Model for Prediction Across Scales (MPAS). Variations in these models—such as their physical formulations, spatial resolution, and computational efficiency—can significantly influence streamflow predictions. In addition, AI/ML based weather prediction approaches (e.g., FourCastNet), are emerging as promising alternatives for improving forecast accuracy.

Objectives

1. Evaluate the impact of different precipitation forecasts from regional NWP models on streamflow simulations using the NGEN framework.
2. Compare hydrological outputs derived from traditional physics-based models and AI-driven approaches.
3. Determine the sensitivity of streamflow predictions to variations in meteorological forcings and identify the most reliable datasets for accurate flood forecasting in the Pascack Brook Watershed.

Study area

The Hackensack River watershed (Figure 1) spans northern New Jersey and southern New York. Within this basin, the Pascack Brook subregion is of particular interest for this study because it contains a stream-gauging station that will be used to calibrate the NextGen framework.

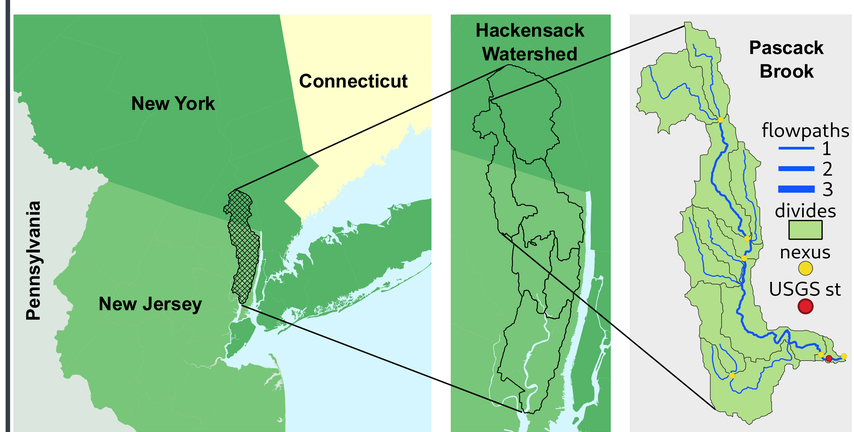

Figure 1. (Left) Map showing the Hackensack River basin between New York and New Jersey; (center) the basin divided into it's subregions; (right) the Pascack Brook subregion highlighted.

Methodology

For this work, we use the NGIAB, taking advantage of two of its core components, Data Preprocess and NGIAB Calibration. The Data Preprocess module streamlines the preparation of all required inputs, while the NGIAB Calibration component provides an intuitive environment for calibrating models such as CFE and NOAH-OWP. Together, these tools provide the inputs, configuration files, and calibrated parameters needed for the correct and efficient execution of the broader NGEN workflow. For the NGEN forcing, we used data from OARC and NWM, while calibration was performed using observations from the USGS stream gauge (ID-01377500).

Calibration and validation

Using NGIAB calibration four experiments, summarized in Table 1, were conducted for calibration and validation, focusing on 2021 (Figure 2), when the remnants of Ida impacted the study area in September.

Table 1. Summary of the experiments conducted to calibrate and validate Pascack Brook

Characteristics	Period	Corr	NSE	KGE	Run time
a) TT: 5Y, W: 1Y, C: 50%, V: 50%, i: 200, D: AORC	Calibration	0.78	0.56	0.78	3:10:09
	Validation	0.74	0.35	0.69	
	Full	0.73	0.39	0.69	
b) TT: 7Y, W: 1Y, C: 80%, V: 20%, i: 500, D: AORC	Calibration	0.83	0.66	0.83	7:37:36
	Validation	0.87	0.74	0.87	
	Full	0.81	0.60	0.75	
c) TT: 7Y, W: 1Y, C: 80%, V: 20%, i: 750, D: AORC	Calibration	0.85	0.69	0.85	11:41:01
	Validation	0.82	0.59	0.78	
	Full	0.79	0.56	0.73	
d) TT: 7Y, W: 1Y, C: 80%, V: 20%, i: 500, D: NWM	Calibration	0.85	0.70	0.85	8:01:45
	Validation	0.84	0.63	0.81	
	Full	0.80	0.58	0.74	

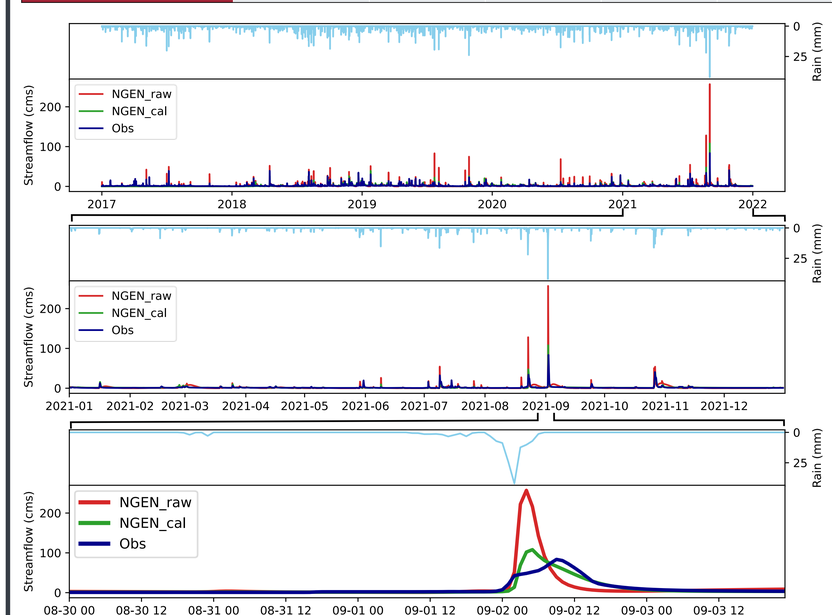

Figure 2. Panels show (upper) the entire calibration time series, (middle) 2021 subset, and (bottom) the Ida range.

Comparison

Once the model was calibrated, forcing data during the passage of the remnants of Hurricane Ida, when it affected our study area, was obtained from several meteorological modeling systems, including the global GFS, the regional WRF and MPAS, and the FCN as an AI-based forecasting approach.

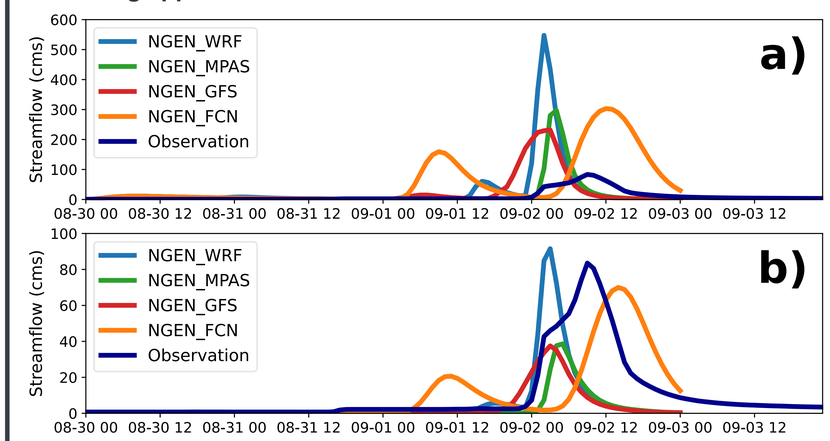

Figure 3. Output of the NGIAB framework when forced with data from different meteorological models: a) results prior to calibration, and b) results after calibration.

Results

- The NGEN framework provides a fast, ready-to-use approach for effective hydrological modeling.
- NGEN-Calib enables rapid calibration and supports experimental workflows to achieve the best match to observations.
- Feature-effects-based calibration is fast but does not always yield the best results. More iterations are not always better; while ~500 iterations performed best here, outcomes also depend on input data.
- Running NGIAB with data from different models shows that, without calibration, all models overestimate the observations. After calibration, the fit improves, though model behavior remains strongly influenced by the input data. The low-resolution global models (GFS, FCN) perform worst, while the regional models generally perform better: MPAS tends to underestimate, and WRF provides best results.

References

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